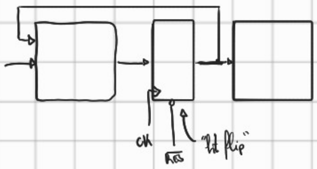
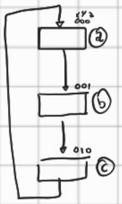
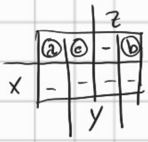
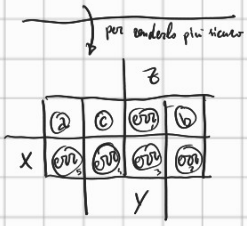
Gestione errori nelle MSF

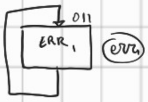

CK
RES
"Flip flop"

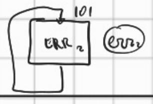

xyz
000
a
001
b
010
c



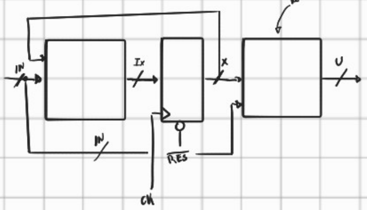
per renderlo più sicuro


z
a
c
err
b
x
err
err
err
err
y


011
ERR1
err1

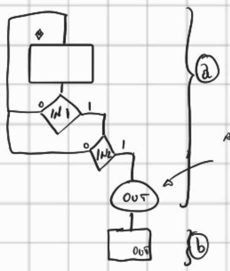

101
ERR2
err2

MSF di Mealy

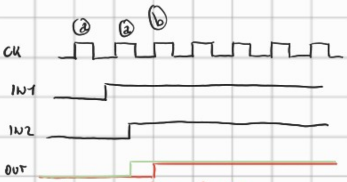

IN
Ix
x
U
IN
RES
CK

Le uscite delle MSF di Mealy dipendono anche dagli ingressi quindi possono essere asincrone

ASM


IN1
0
1
IN2
0
1
OUT
OUT
a
b

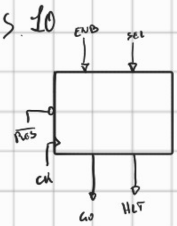
anticipazione di out=1 rispetto a ck



con la MSF di Mealy (asincrono)

fare la MSF di Moore

ES. 10

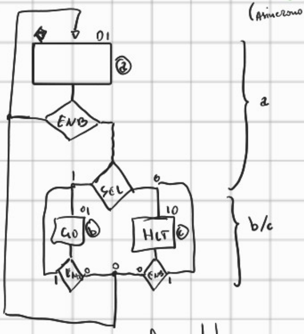

ENB
SEL
RES
CK
GO
HLT

La MSF attende ENB=1

SEL=1 → GO=1

SEL=0 → HLT

Le uscite rimangono attive per tutto il tempo in cui ENB=1

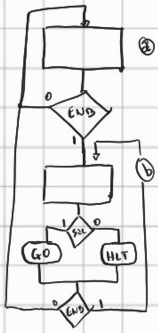

01
a
ENB
SEL
1
0
01
10
GO
b
HLT
c
ENB
ENB
1
0
0
0
1
a
b/c

M1 (macchina)

ha 3 stati

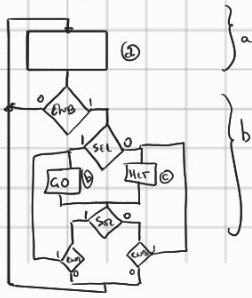
Mealy ha due stati

ha 2 stati


a
0
ENB
1
b
1
SEL
0
GO
HLT
0
ENB
1

M3

può cambiare SEL nel frattempo


a
0
ENB
1
1
SEL
0
GO
b
HLT
c
1
SEL
0
1
ENB
ENB
0
0
a
b/c

M2

M1 ≠ M2 ~ M3

Mappa degli Stati

	X
ⓐ	ⓑ

Rete comb. stato successivo

FFD:

D_x = ENB

	X
EN_B	EN_B

Rete comb. uscite:

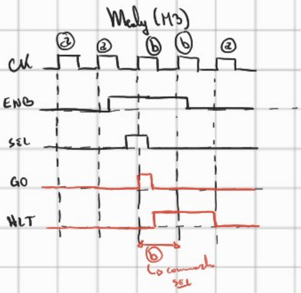
stato in cui sono

	X
0	SEL

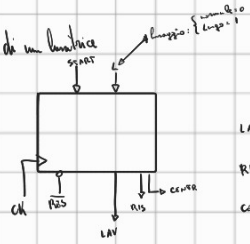
	X
0	$\overline{SEL}$

GO = SEL (X)   variabili riportate   HLT = $\overline{SEL}$ (x)

Mealy (M3)


ⓐ
ⓐ
ⓑ
ⓑ
ⓐ
CK
ENB
SEL
GO
HLT
ⓑ
↳ commuta SEL

Controllo digitale di una lavatrice

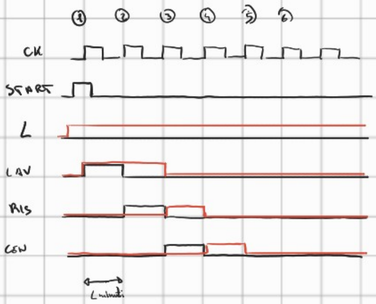

lavaggio: { normale: 0, lungo: 1
START
L
CK
$\overline{RES}$
LAV
RIS
CENTR

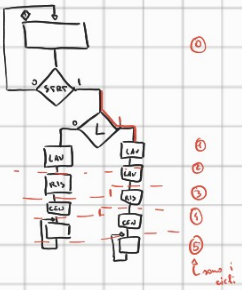
LAV=1 lavaggio

RIS=1 risciacquo

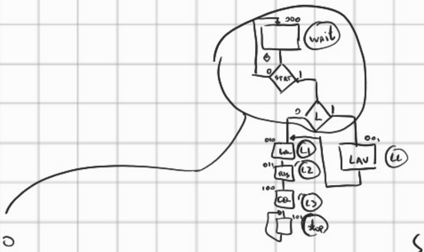
CENTR=1 centrifuga

START 0→1 inizio lavaggio


①
②
③
④
⑤
⑥
CK
START
L
LAV
RIS
CEN
1 minuto


0
START
1
0
L
1
LAV
RIS
CEN
LAV
LAV
RIS
CEN
⓪
①
②
③
④
⑤
sono i cicli

MSF "ottimizzata" v2.0


000
wait
0
START
1
0
L
1
010
LAV
L1
011
RIS
L2
100
CEN
L3
101
stop
001
LAV
LL

X 0→0

Y 0 →0, →1 START·$\overline{L}$

Z: 0 →0, ↳1 START L

Reg a 3 bit ↳ anche se poi non cambia in corso

Trans	J	K
0→0	0	–
0→1	1	–
1→0	–	1
1→1	–	0

Sintesi con il JK

			Z	
	wait	L1	L2	LL
X	L3	–	–	stop
		Y		

Mappe. CSS

			Z	
	0	0	1	0
X	–	–	–	–
		Y		

$J_x = YZ$

			Z	
	–	–	–	–
X	0	–	–	0
		Y		

$K_x = 0$

			Z	
	START·$\overline{L}$	–	–	1
X	0	–	–	0
		Y		

$J_y = \overline{X}Z + START \cdot \overline{L} (\overline{XZ})$

			Z	
	–	0	1	–
X	–	–	–	–
		Y		

$K_y = Z$

			Z	
	START·L	1	–	–
X	1	–	–	–
		Y		

$J_z = 1 + START\,L$

			Z	
	–	–	1	1
X	–	–	–	0
		Y		

$K_z = \overline{X}$